Hypothesis one sample t-tests

1

a $t = -3.35$

b p-value $= P(T \leq -3.35), T \sim t_{45-1}$

c $p = 0.0008$

d The low p-value indicates H_0 is likely to be rejected.

2

a $t = 2.21$

b p-value $= P(T \geq 2.21), T \sim t_{60-1}$

c $p = 0.0155$

d H_0 will be rejected.

3

a $t = -0.40$

b $p\text{-value} = P(T > 0.40)$ or $p\text{-value} = P(T < -0.40), T \sim t_{30-1}$.

c $p = 0.6921$

d H_0 will not be rejected.

Applications

4

a $H_0 : \mu = 0.06, H_1 : \mu > 0.06$

b $t = 0.0844$

c $p = 0.4665$

d No, the proportion of 6% will not be rejected because the p-value is more than 10%, therefore the sample statistics are realistic at this significance level.

5

a $H_0 : \mu = 52, H_1 : \mu > 52$

b $t = 1.78$

c $p = 0.0391$

d H_0 will be rejected.

6

a $H_0 : \mu = 34, H_1 : \mu < 34$

b $t = -1.33$

c $p = 0.0948$

d No, the original mean level of $\mu = 34$ ppm of pollutants will not be rejected because the p-value is more than 5%, therefore the sample statistics are realistic for this significance level.

7

a $H_0 : \mu = 1135, H_1 : \mu \neq 1135$

b $t = -1.84$

c $p = 0.0814$

d Yes, the mean cost of $\mu = \$1135$ will be rejected because the p-value is less than 10%, therefore the sample mean is improbable for $\mu = \$1135$ at this significance level.

8

a $H_0 : \mu = 300, H_1 : \mu \neq 300$

b $p = 0.15$

c $\overline{x} = 299.08$ ml

d No, there is not a problem and the mean amount of $\mu = 300$ ml will not be rejected because the p-value is more than 1%. The sample mean is probable for $\mu = 300$ ml at this significance level.